15.095: Machine Learning Under a Modern Optimization Lens **Fall 2026**

Homework 1: Due September 23

Hand in: **pdf** upload to Gradescope. Please append any Julia code **at the end** of the whole pdf.

Note: this homework covers lectures 1–3 and recitations 1–2. We recommend attempting questions after the relevant content has been covered in class. Total: 100 points.

Question 1: True/False (30 points)

Please classify the following statements as true or false and justify your answer. We will assign 1 point for correctly classifying the answer, and 4 points for the validity of the justification/counterexample.

(a) (5 points) Consider the linear optimization problem

$$\begin{aligned} \max_{x,y\in\mathbb{R}} \quad & x + 2y \\ \text{s.t.} \quad & x \le y, \\ & -x + y \le -1, \\ & x,\, y \ge 0. \end{aligned}$$

The dual of this problem is unbounded.

(b) (5 points) The Elastic Net problem

$$\min_{\boldsymbol{\beta}} \ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2 + \lambda\|\boldsymbol{\beta}\|_1 + \mu\|\boldsymbol{\beta}\|_2$$

simultaneously protects against global perturbations of $\mathbf{X}$ bounded by λ and feature-wise perturbations bounded by μ.

(c) (5 points) Let $M \ge 1$ be a known constant. Consider the optimization problem

$$\begin{aligned} \min_{\boldsymbol{x},\,\boldsymbol{y}\in\mathbb{R}^n} \quad & \|\boldsymbol{x} - \boldsymbol{y}\|_1 \\ \text{s.t.} \quad & \|\boldsymbol{x}\|_1 + \|\boldsymbol{y}\|_1 \ge 2, \\ & \|\boldsymbol{y}\|_0 \le 5, \\ & \|\boldsymbol{x}\|_\infty \le M, \\ & \|\boldsymbol{y}\|_\infty \le M. \end{aligned}$$

This problem can be reformulated as a mixed-integer linear optimization problem.

(d) (5 points) Consider adding noise to the labels, with $\boldsymbol{\delta} \in \mathcal{U} = \{\boldsymbol{\delta} \in \mathbb{R}^n : \|\boldsymbol{\delta}\|_2 \le \lambda\}$. Then

$$\min_{\boldsymbol{\beta}} \max_{\boldsymbol{\delta}\in\mathcal{U}} \ \|\mathbf{y} + \boldsymbol{\delta} - \mathbf{X}\boldsymbol{\beta}\|_2 \ = \ \min_{\boldsymbol{\beta}} \ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2 + \lambda\|\boldsymbol{\beta}\|_2.$$

(e) (5 points) Recall the big-M formulation of sparse regression,

$$\begin{aligned} \min_{\boldsymbol{\beta}\in\mathbb{R}^p,\ \boldsymbol{z}\in\{0,1\}^p} \quad & \tfrac{1}{2}\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|_2^2 + \tfrac{1}{2\gamma}\|\boldsymbol{\beta}\|_2^2 \\ \text{s.t.} \quad & -Mz_j \le \beta_j \le Mz_j, \quad \forall j\in[p], \\ & \sum_{j=1}^p z_j \le k. \end{aligned}$$

To scale sparse regression to instances with n and p in the 100,000s, we run the cutting-plane algorithm directly on this formulation.

(f) (5 points) Recall the Lasso problem,

$$\min_{\boldsymbol{\beta}} \ \|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|_2 + \lambda\|\boldsymbol{\beta}\|_1.$$

Lasso is not equivalent to sparse regression.

Question 2: What Is That Sublet Actually Worth? (40 points)

Every spring, hundreds of students look for a summer sublet in Cambridge and Somerville, and every spring they have no idea whether \$2,400 for a one-bedroom near Central Square is a reasonable price or a fleecing. You decide to build a model that predicts the typical advertised monthly rent from a listing's features, and to publish it so incoming students have a benchmark to compare an asking price against before they sign.

You scrape n listings. For listing i you record $\mathbf{x}_i \in \mathbb{R}^p$, a vector of p features, and the advertised monthly rent $y_i \in \mathbb{R}$. Stack these into $\mathbf{X} \in \mathbb{R}^{n\times p}$ and $\mathbf{y} \in \mathbb{R}^n$, and let $\boldsymbol{\beta} \in \mathbb{R}^p$ be the regression coefficients you are fitting.

The advertised rent y_i is exactly what it says on the page, so $\mathbf{y}$ is reliable. The feature matrix $\mathbf{X}$ is not, and it can fail in two quite different ways. Individual recorded values may each be somewhat inaccurate: square footage is rounded, walking times are estimated by whoever wrote the listing, and neither error has a consistent direction. Or the recorded feature vector for a single listing may be unreliable as a whole, because one poster has substantially reshaped several attributes at once. Such a poster has only so much nerve, though: the more fields they stretch, the more obvious the listing looks, so what is really capped is the total distortion spent on that listing, not the distortion in any one field.

Write $\boldsymbol{\Delta} \in \mathbb{R}^{n\times p}$ for a perturbation of $\mathbf{X}$, with $\boldsymbol{\Delta}_j \in \mathbb{R}^n$ its column j and $\boldsymbol{\delta}_i \in \mathbb{R}^p$ its row i. The two failures give two sets:

$$\mathcal{A} = \{\boldsymbol{\Delta} \mid \|\boldsymbol{\Delta}_j\|_\infty \le \lambda \quad \forall j\in[p]\}, \qquad \mathcal{B} = \{\boldsymbol{\Delta} \mid \|\boldsymbol{\delta}_i\|_1 \le \rho \quad \forall i\in[n]\},$$

with radii $\lambda > 0$ and $\rho > 0$, so $\mathcal{A}$ caps how far any single recorded number can be off, and $\mathcal{B}$ gives each listing a budget of distortion to spend however it likes across its features.

Recall the induced-norm sets from lecture,

$$\mathcal{U}_{(r,q)} = \left\{\boldsymbol{\Delta} \;\middle|\; \|\boldsymbol{\Delta}\|_{(r,q)} \le \varrho\right\}, \qquad \|\boldsymbol{\Delta}\|_{(r,q)} := \max_{\boldsymbol{x}\neq\mathbf{0}} \frac{\|\boldsymbol{\Delta}\boldsymbol{x}\|_q}{\|\boldsymbol{x}\|_r},$$

and recall that $\mathcal{U}_{(1,2)}$ turned out to be exactly a per-column budget, because $\|\boldsymbol{\Delta}\|_{(1,2)} = \max_j \|\boldsymbol{\Delta}_j\|_2$. Nothing in that argument used the 2: the same reasoning gives, for every q,

$$\|\boldsymbol{\Delta}\|_{(1,q)} = \max_{j\in[p]} \|\boldsymbol{\Delta}_j\|_q, \tag{$\star$}$$

so a first index of 1 means "budget each column on its own, measuring it in ℓ_q". You may use ($\star$) without proof. Taking $q = \infty$ places the first of our two sets: $\mathcal{A} = \mathcal{U}_{(1,\infty)}$ with radius λ. Note what that set actually is — capping every column in ℓ_∞ is the same as capping every entry, so $\mathcal{A}$ is simply the box $|\Delta_{ij}| \le \lambda$. However, ($\star$) is not enough to write the second set.

Part A (25 points)

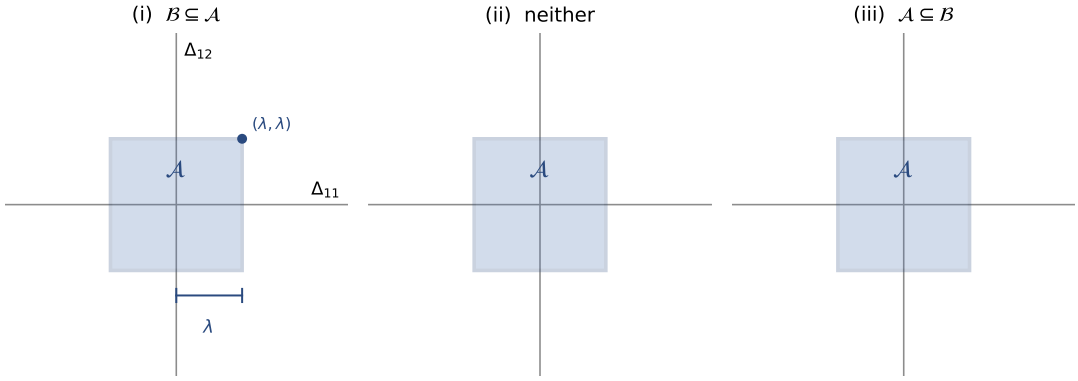
This part is about what $\mathcal{A}$ and $\mathcal{B}$ are as regions of $\mathbb{R}^{n\times p}$.

(a) (15 points) The set $\mathcal{B}$ is also an induced-norm ball: $\mathcal{B} = \mathcal{U}_{(\infty,\infty)}$ with radius ρ. Verify this, by proving $\|\mathbf{\Delta}\|_{(\infty,\infty)} = \max_i \|\boldsymbol{\delta}_i\|_1$; both inequalities are needed. Then notice the pairing this exposes: the budget on a row is measured in ℓ_1, yet the set is $\mathcal{U}_{(\infty,\infty)}$. Explain where the switch from 1 to ∞ comes from, and say which row budget would instead give a first index of 2.

Hints. (i) The ℓ_∞ norm of $\mathbf{\Delta}\boldsymbol{x}$ separates the rows: $\|\mathbf{\Delta}\boldsymbol{x}\|_\infty = \max_{i\in[n]} |\boldsymbol{\delta}_i^\top \boldsymbol{x}|$. That leaves two maximizations stacked on each other; consider whether they can be taken in either order. (ii) Each inner problem should then match the dual norm taken with respect to ℓ_∞. Evaluate it from the definition: for the upper bound, expand $\boldsymbol{\delta}_i^\top \boldsymbol{x}$ as a sum and use $|x_j| \le \|\boldsymbol{x}\|_\infty$; for the matching lower bound, the signs of the entries of $\boldsymbol{\delta}_i$ tell you which single $\boldsymbol{x}$ to try.

(b) (10 points) Take the smallest case in which both sets are visible: one listing and two features, $n = 1$ and $p = 2$, so that a perturbation is a single row $\boldsymbol{\delta}_1 = (\Delta_{11}, \Delta_{12}) \in \mathbb{R}^2$ and each set is a region of the plane.

Because ρ and λ are independent, the two sets can sit in three ways: $\mathcal{B} \subseteq \mathcal{A}$, neither one contained in the other, or $\mathcal{A} \subseteq \mathcal{B}$. Below are three copies of $\mathcal{A}$, one per case.



On each copy, sketch a $\mathcal{B}$ that realizes the case named above it, and state the condition on λ and ρ under which that case occurs.

(c) (+10 points extra credit) Now step back out of the plane: keep one listing, but let the number of features be a general p rather than 2. How do your two containment conditions, (i) and (iii), change as p grows?

Part B (15 points)

(15 points) You now fit the model, defending against both failures at once. Because one student badly misled by your published rule is a worse outcome than a small average error, you measure fit by $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_\infty$. And because students have to read the rule, it may use at most k of the p features.

From lecture you may quote without proof that

$$\max_{\boldsymbol{\Delta}\in\mathcal{U}_{(r,q)}} \|\mathbf{y}-(\mathbf{X}+\boldsymbol{\Delta})\boldsymbol{\beta}\|_q = \|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|_q + \varrho\|\boldsymbol{\beta}\|_r,$$

and that perturbations drawn from two such sets and added to $\mathbf{X}$ together contribute both penalties.

Formulate

$$\min_{\boldsymbol{\beta}} \max_{\boldsymbol{\Delta}\in\mathcal{A},\,\boldsymbol{\Delta}'\in\mathcal{B}} \|\mathbf{y}-(\mathbf{X}+\boldsymbol{\Delta}+\boldsymbol{\Delta}')\boldsymbol{\beta}\|_\infty \qquad \text{s.t.} \quad \|\boldsymbol{\beta}\|_0 \le k$$

as a mixed-integer optimization problem. You may introduce auxiliary continuous or binary variables, and M_j denotes a big-M constant for coefficient j. State what each variable does.

Question 3: Coding the Sublet Rule (30 points)

This question continues the Cambridge / Somerville sublet story from Question 2. You will implement the mixed-integer formulation you derived in Part B and run it on a small listing dataset.

Use the accompanying Jupyter notebook `hw1-starter.ipynb` and the dataset `sublets.csv`. Place both files in the same directory, fill in the marked code cells, run the notebook, and report the numerical outputs below. You may assume access to JuMP and Gurobi (as in recitation).

The file `sublets.csv` has $n = 30$ listings and $p = 7$ features. The response column is `rent` (monthly asking price in dollars). The feature columns are

`sqft`, `beds`, `baths`, `walk_t`, `laundry`, `parking`, `floor`.

Here `walk_t` is walking time in minutes to the nearest T stop, and `laundry` / `parking` are binary amenities. The starter notebook centers $\mathbf{y}$ and standardizes the columns of $\mathbf{X}$ (so no intercept is needed). Throughout, take the big-M constant $M = 10^3$ for every feature.

(a) (15 points) In the starter notebook, complete the function `solve_robust_sparse` that builds and solves this mixed-integer optimization problem with JuMP / Gurobi. Then run it with

$$\lambda = 0.25, \qquad \rho = 0.5, \qquad k = 3.$$

Report the optimal objective value, the optimal residual level $\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\|_\infty$, the selected features (those with non-negative weights, i.e., $|\beta_j| > 10^{-6}$), and their corresponding coefficients β_j.

(b) (10 points) Keep $k = 3$ and compare the three uncertainty regimes

Case	(λ, ρ)	Interpretation
$\mathcal{A}$ only	(0.25, 0)	per-entry feature noise
$\mathcal{B}$ only	(0, 1)	per-listing fabrication budget
both	(0.25, 0.5)	same as in (a)

For each case, report the selected features and coefficients. In one or two sentences, explain how the ℓ_1 penalty (from $\mathcal{A}$) and the ℓ_∞ penalty (from $\mathcal{B}$) shape the coefficients differently.

(c) (5 points) Return to $\lambda = 0.25$, $\rho = 0.5$, and compare the sparse model $k = 3$ to the dense model $k = 7$ (i.e. no effective cardinality restriction). Report u in each case and the nonzero coefficients for $k = 7$. In one sentence, say what a student using your published rule gains or loses by insisting on at most three features.